

PARKTOWN BOYS' HIGH SCHOOL



Department of Mathematics

Preliminary Examination Paper 2

26 September 2023

Subject	Mathematics
Grade	12
Marks	150
Time	3 hours

Examiner

Mrs S Rowell

Moderator

Mrs H Gouws

INSTRUCTIONS AND INFORMATION

1. This paper consists of 11 pages and a formula sheet.
 - You may detach the formula sheet for your own use.
2. There are 9 questions.
3. Answer all questions in the answer book provided. Additional space is provided at the end of the answer book.
4. Round all answers off to TWO decimals unless indicated otherwise in a particular question.
5. Approved non-graphical and non-programmable calculators may be used.
6. An answer alone will not necessarily be awarded full marks.
7. Look at the mark allocation as an indication of the length of the solution.
8. Write neatly and legibly.

QUESTION 1

The table below shows the age and the number of international caps of 10 selected Springbok rugby players for the Rugby World Championship 2023.

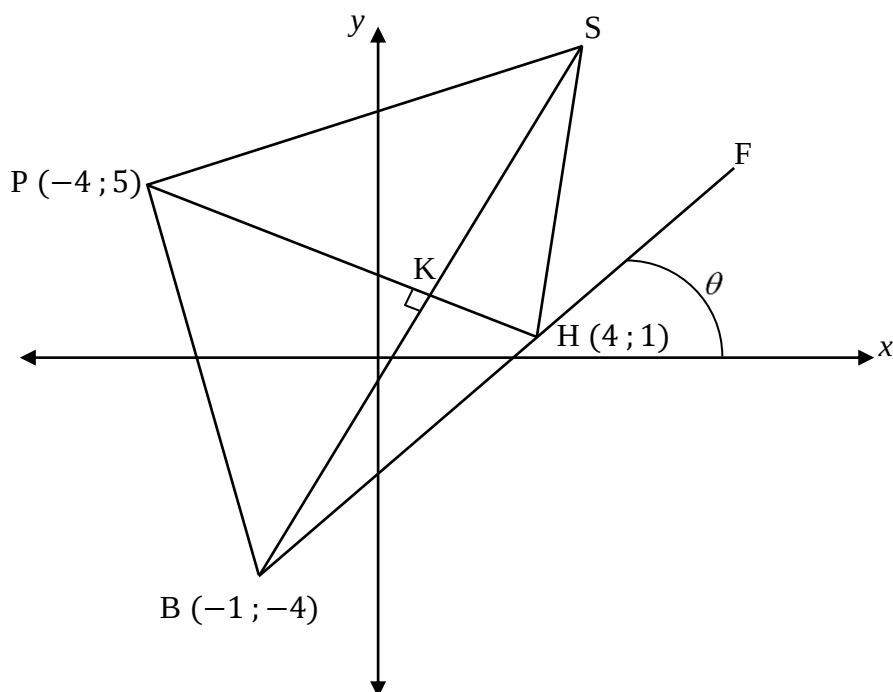
Player	Age in years (x)	Number of caps (y)
Canan Moodie	20	3
RG Snyman	28	23
Jesse Kriel	29	57
Damian De Allende	31	61
Eben Etzebeth	31	112
Pieter-Steph Du Toit	31	61
Siya Kolisi	32	74
Frans Malherbe	32	61
Trevor Nyakane	34	57
Willie Le Roux	34	87

- 1.1 Consider the number of caps of the players.
 - 1.1.1 Calculate the mean and the standard deviation. (2)
 - 1.1.2 How many players lie below one standard deviation from the mean? (2)
- 1.2 Consider the ages of the players.
 - 1.2.1 Draw a box-and-whisker diagram to represent the ages of the players. (4)
 - 1.2.2 Comment on the distribution of the ages of the players. (1)
- 1.3 Consider the relationship between the age and the number of caps of the players.
 - 1.3.1 Determine the equation of the least squares regression line. (3)
 - 1.3.2 Calculate the correlation co-efficient. (1)
 - 1.3.3 Comment on the relationship between the age and the number of caps of the players. (2)
 - 1.3.4 Predict the number of caps a Springbok player will have at the age of 33 years. (2)
 - 1.3.5 How reliable is your prediction in 1.3.4? Motivate your answer. (2)

[19]

QUESTION 2

In the diagram below, $P(-4; 5)$, $B(-1; -4)$, $H(4; 1)$ and S are the vertices of a quadrilateral. BH is produced to F . K is the point of intersection of the diagonals of $PBHS$. $PH \perp BS$. θ is the angle of inclination of line BF .

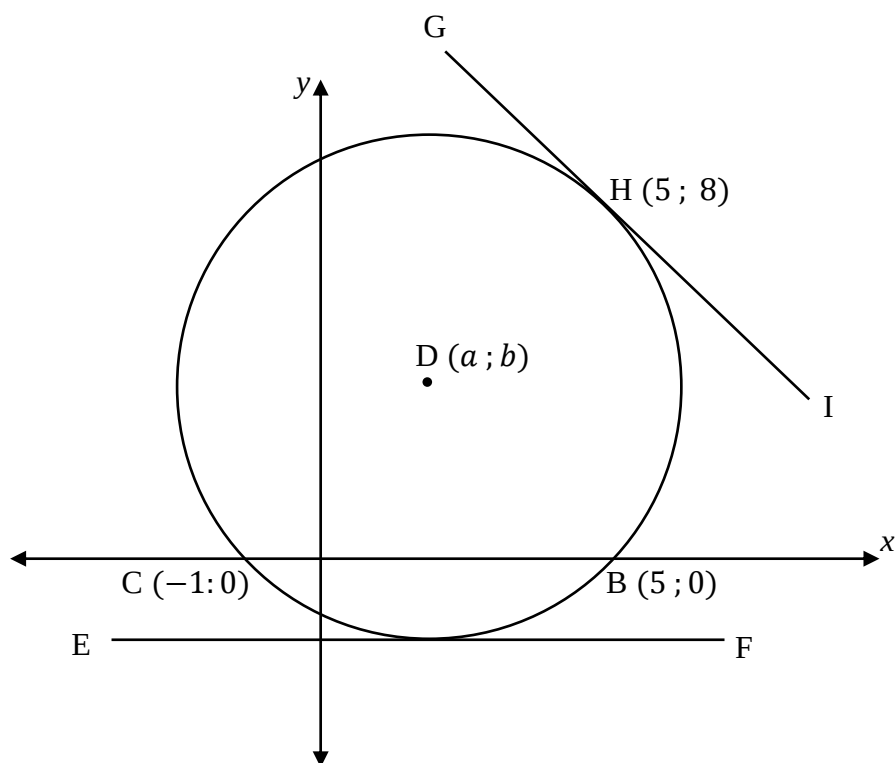


- 2.1 Determine the gradient of PH . (2)
- 2.2 Determine the equation of BS . (3)
- 2.3 Show that the co-ordinates of K are $(2; 2)$. (4)
- 2.4 Calculate the size of $\hat{PHF}$. (5)
- 2.5 Calculate the area of $\triangle PBH$. (4)
- 2.6 If there is a point A such that $PBAH$ is a parallelogram, determine the coordinates of A . (4)

[22]

QUESTION 3

In the diagram below, the circle has a centre at $D(a; b)$. $C(-1; 0)$ and $B(5; 0)$ are points on the x -axis. EF is a tangent to the circle parallel to the x -axis. GI is a tangent to the circle at $H(5; 8)$.

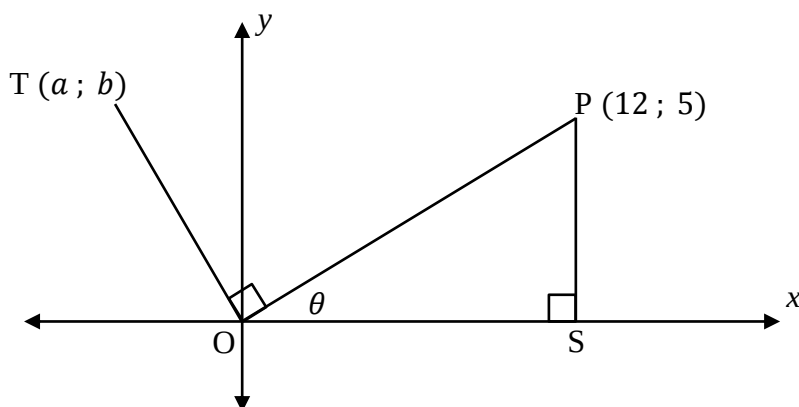


- 3.1 Show that $a = 2$ and $b = 4$. (2)
- 3.2 Determine the equation of the circle. (3)
- 3.3 Determine the equation of EF . (2)
- 3.4 Show that C , D and H are collinear. (3)
- 3.5 Determine the equation of the tangent GI . (3)
- 3.6 If a circle with centre $N(p; 4)$ and a radius of 4 units intersects the circle BCH at two distinct points, determine the possible values of p . (5)

[18]

QUESTION 4

- 4.1 In the diagram below, P is the point (12 ; 5) and T (a ; b). $OT \perp OP$, $PS \perp x\text{-axis}$ and $\hat{POS} = \theta$.



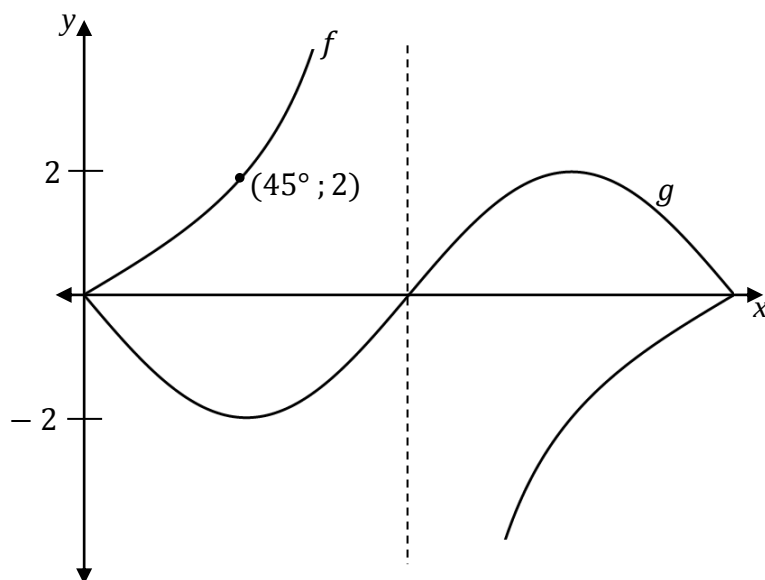
Without the use of a calculator, determine the value of:

- 4.1.1 $\tan \theta$ (1)
- 4.1.2 $\sin \theta$ (2)
- 4.1.3 a , if $TO = 19,5$ units (3)
- 4.2 Simplify the following to a single trigonometric ratio:
- $$\frac{\cos(180^\circ + x) \cdot \tan(540^\circ - x) \cdot \sin^2(90^\circ + x)}{\sin(180^\circ - x)} - \sin^2 x$$
- (6)
- 4.3 Determine the value of $\cos 105^\circ$ without the use of a calculator. (4)
- 4.4 If $\sin 31^\circ \cdot \cos 22^\circ + \sin 22^\circ \cdot \cos 31^\circ = k$, determine the value of $\tan 53^\circ$ in terms of k . (4)
- 4.5 Prove the identity $\frac{\sin 3\theta}{\sin \theta} = 4\cos^2 \theta - 1$. (5)
- 4.6 Determine the general solution to $2\cos^2 x = 3\sin x$. (5)

[30]

QUESTION 5

Sketched below are the functions $f(x) = a \tan x$ and $g(x) = b \sin kx$ for $x \in [0^\circ; 180^\circ]$.
 $f(45^\circ) = 2$.

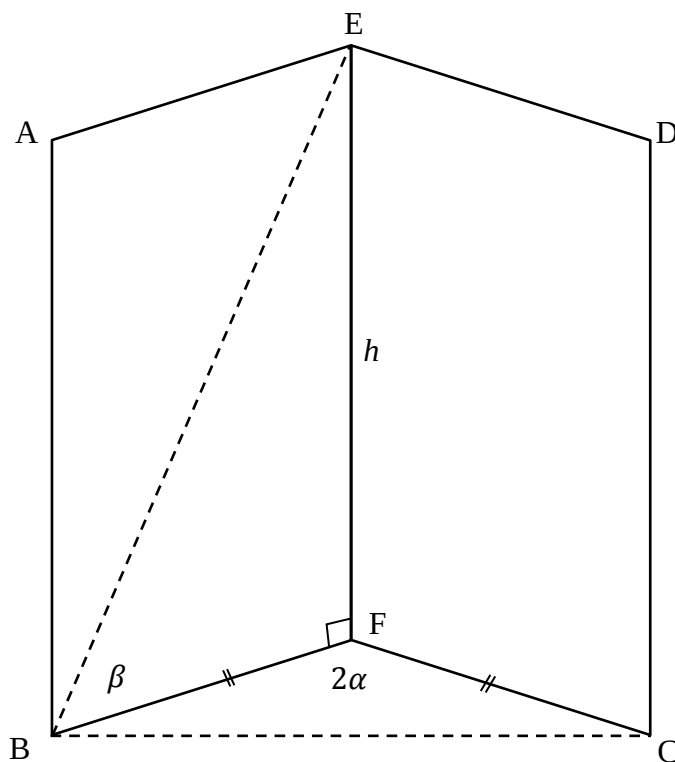


- 5.1 Write down the values of a , b and k . (3)
- 5.2 For which values of x is $f(x) \geq g(x)$? (3)
- 5.3 A new graph h is obtained by reflecting f over the x -axis and shifting it 2 units up. Write down the equation of h in the form $h(x) = \dots$ (2)
- 5.4 Write down the equation of the asymptote of h . (1)
- 5.5 Give an equation to represent g using cosine rather than sine. Write the equation in the form $y = \dots$ (1)

[10]

QUESTION 6

The pamphlet for the Parktown Boys' High School Cultural Evening stands upright on a table as seen in the figure below. It is opened at an angle of 2α . ABFE and EFCD are identical rectangles. BE is the diagonal of ABFE and $\hat{EBF} = \beta$. The height of the pamphlet EF is h units.



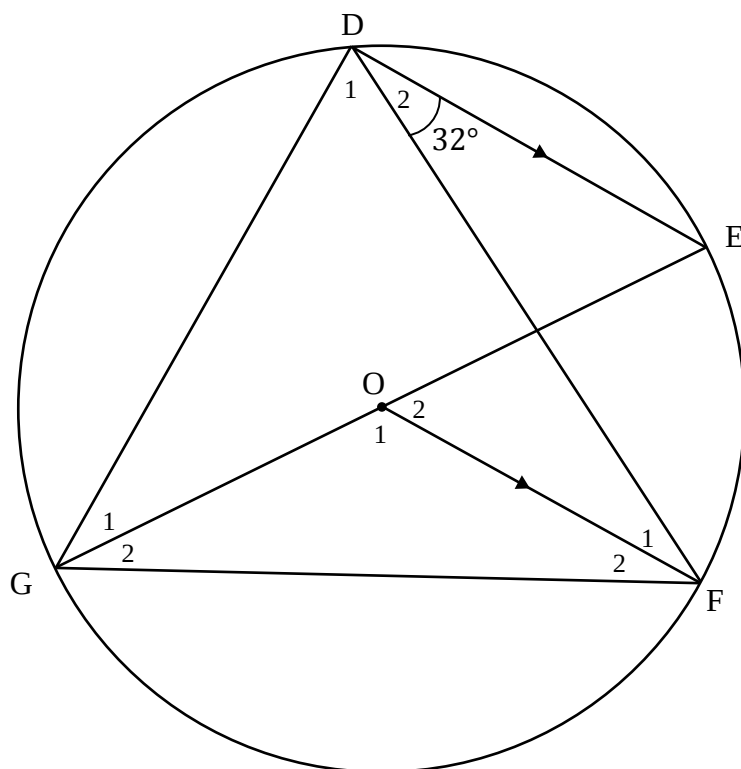
6.1 Show that $BC = \frac{2h \sin \alpha}{\tan \beta}$. (7)

6.2 Determine the height of the pamphlet if $BC = 25$ cm, $\alpha = 50^\circ$ and $\beta = 55^\circ$. (2)

[9]

QUESTION 7

In the diagram below, O is the centre of the circle and D, E, F and G are on the circumference so that GE is a diameter. $\hat{D}_2 = 32^\circ$ and $DE \parallel OF$.



Determine, with reasons, the size of:

7.1 $\hat{D}_1$ (2)

7.2 $\hat{O}_1$ (2)

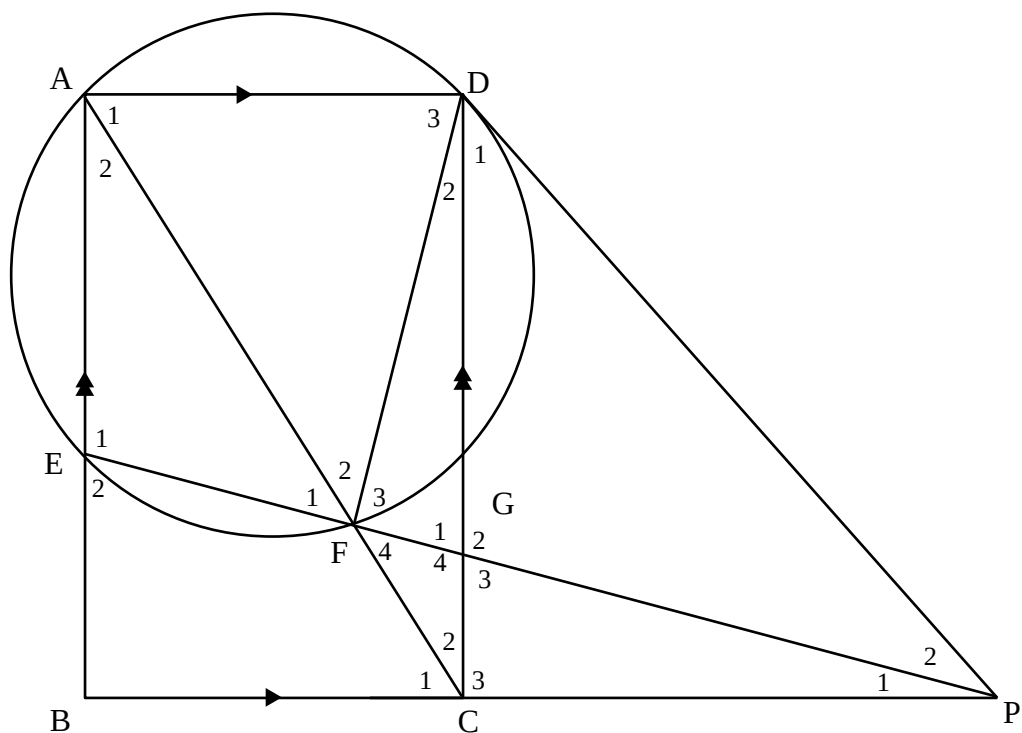
7.3 $\hat{F}_2$ (2)

7.4 $\hat{E}$ (3)

[9]

QUESTION 8

ABCD is a rectangle with diagonal AC. Circle AEFD cuts AB at E and AC at F. BC is produced to P. PD and FD are joined. DC and EP intersect at G. $AD \parallel BCP$ and $AEB \parallel DGC$. You may not assume that DP is a tangent to the circle.



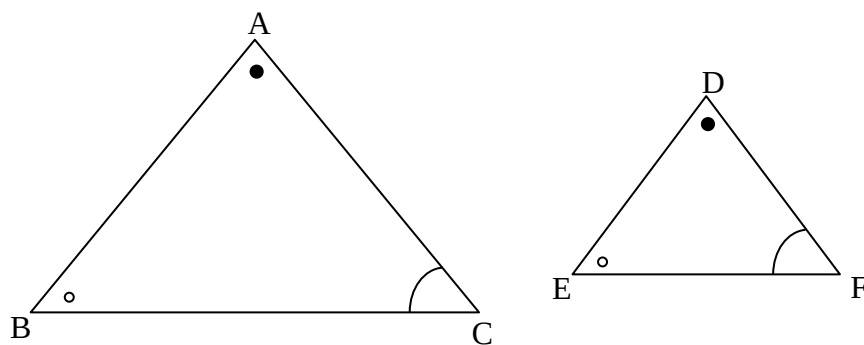
- 8.1 Prove that $EP \perp FD$. (2)
- 8.2 Prove that DFCP is a cyclic quadrilateral. (2)
- 8.3 Prove that DP is a tangent to the circle AEFD. (4)
- 8.4 It is further given that $4BC = 5CP$ and $EF:FG = 3:1$. Determine the length of FG if $EP = 36$ units. (4)

[12]

QUESTION 9

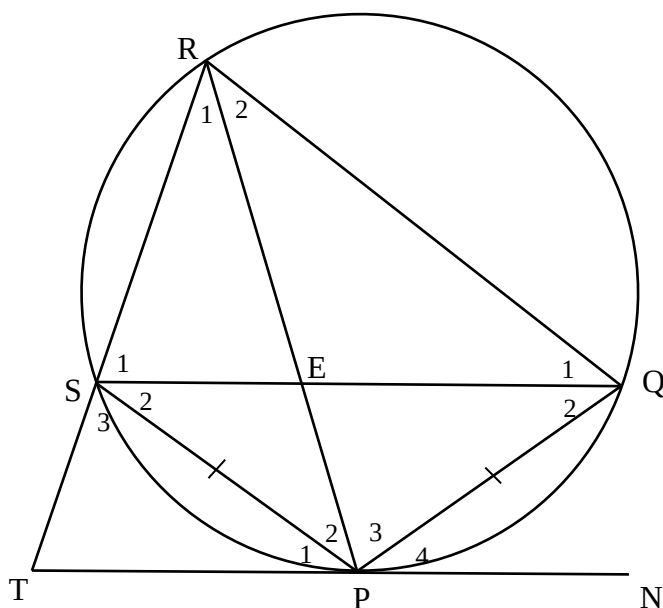
- 9.1 Given below are two triangles where $\hat{A} = \hat{D}$, $\hat{B} = \hat{E}$ and $\hat{C} = \hat{F}$.
Use the sketch in the answer book to prove the theorem that states:

“The corresponding sides of two equiangular triangles are in the same proportion and therefore the triangles are similar.”



(6)

- 9.2 Circle PQRS has a tangent TN meeting the circle at P. $SP = PQ$.



Prove that:

9.2.1 $TN \parallel SQ$ (4)

9.2.2 $\triangle TSP \parallel \triangle TPR$ (4)

9.2.3 $SP^2 = TS \times QR$ (5)

9.2.4 $PR^2 = TR \times QR$ (2)

[21]